

OIS-012 — Service Contracts

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-011
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the constitutional implementation contract model.

Every executable capability within Organodynamics SHALL be exposed through an explicit Service Contract.

A Service Contract defines operational behavior.

It SHALL NOT define constitutional meaning.

Every implementation SHALL preserve constitutional behavior by honoring Service Contracts.

1. Implementation Objective

The objective of Service Contracts is to guarantee interoperable behavior among independently developed implementations.

Every Contract SHALL preserve:

Behavior.

Identity.

Historical continuity.

Operational predictability.

Replaceability.

Contracts SHALL remain implementation-neutral.

2. Definition

A Service Contract is the executable agreement governing one operational responsibility.

Every Contract SHALL define:

Purpose.

Inputs.

Outputs.

Observable behavior.

Failure behavior.

Constitutional invariants.

Compatibility guarantees.

3. Contract Identity

Every Service Contract SHALL possess:

Canonical identity.

Version.

Applicable specifications.

Historical lineage.

Compatibility relationships.

Contracts SHALL remain independently referenceable.

4. Inputs

Every Contract SHALL explicitly define:

Accepted constitutional constructs.

Accepted operational metadata.

Validation rules.

Required context.

Optional context.

No hidden input SHALL influence constitutional behavior.

5. Outputs

Every Contract SHALL explicitly define:

Produced constitutional constructs.

Produced Events.

Operational metadata.

Observable side effects.

Expected projections.

Outputs SHALL remain deterministic whenever constitutionally required.

6. Failure Behavior

Every Contract SHALL explicitly define failure behavior.

Failure SHALL distinguish between:

Operational failure.

Architectural violation.

Constitutional violation.

Recovery expectations.

Failure SHALL remain observable.

7. Constitutional Invariants

Every Service Contract SHALL declare the constitutional invariants it preserves.

Examples include:

Identity continuity.

Event immutability.

Historical reconstruction.

Relationship preservation.

Schema compatibility.

Violation SHALL constitute implementation non-conformance.

8. Compatibility

Contracts SHALL distinguish:

Backward-compatible changes.

Forward-compatible changes.

Breaking changes.

Experimental extensions.

Compatibility SHALL remain explicit.

9. Verification

Verification SHALL demonstrate:

Behavioral correctness.

Invariant preservation.

Failure correctness.

Compatibility.

Historical continuity.

Equivalent constitutional behavior.

10. Service Contract Test

A Service Contract satisfies this specification when independently developed implementations can invoke one another's services while preserving equivalent constitutional behavior without prior implementation-specific coordination.

Conformance

An implementation conforms to this specification when every operational capability is governed by explicit Service Contracts whose behavior, invariants, compatibility, and failure semantics remain constitutionally equivalent across implementations.

Implementation Principle

Code changes.

Protocols evolve.

Infrastructure is replaced.

Contracts remain.

The purpose of a Service Contract is therefore not merely to define communication,

but to preserve constitutional expectations between independent implementations,
ensuring that engineering diversity never compromises constitutional integrity.

End of Specification OIS-012